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Task :

1 . Console Application for managing the employee records using the c# language

Solution:

Employee :

```
using System;
namespace ConsoleApp6
{
    internal class Employee
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public double Salary { get; set; }

        public virtual string GetDetails()
        {
            return $"ID: {Id}, Name: {Name}, Salary: {Salary}";
        }

        public override string ToString()
        {
            return GetDetails();
        }
    }
}
```

Manager :

```
using System;

namespace ConsoleApp6
{
    internal class Manager:Employee
    {
        public string Department { get; set; }
    }
}
```

```

        public override string GetDetails()
        {
            return base.GetDetails() + $", Department: {Department}";
        }

        public override string ToString()
        {
            return GetDetails();
        }
    }
}

```

Intern:

```

using System;

namespace ConsoleApp6
{
    internal class Intern : Employee
    {
        public int DurationInMonths { get; set; }

        public override string GetDetails()
        {
            return base.GetDetails() + $", Internship Duration: {DurationInMonths} months";
        }

        public override string ToString()
        {
            return GetDetails();
        }
    }
}

```

Employee Interface :

```

using System;

namespace ConsoleApp6.Repo
{
    public interface IEmployeeRepo<T>
    {
        void AddEmployee(T emp);
    }
}

```

```

        List<T> GetAllEmployees();
    }
}

```

Interface Implementation :

using System;

```

namespace ConsoleApp6.Repo
{
    public class EmployeeRepo<T> : IEmployeeRepo<T>
    {
        private List<T> employees = new List<T>();

        public void AddEmployee(T emp)
        {
            employees.Add(emp);
        }

        public List<T> GetAllEmployees()
        {
            return employees;
        }
    }
}

```

Main Program:

```

using System;
using ConsoleApp6;
using ConsoleApp6.Repo;

```

```

namespace Day2
{
    class program
    {
        static void Main(string[] args)
        {
            Employee emp1 = new Employee { Id = 1, Name = "Ravi", Salary = 50000 };
            Employee emp2 = new Employee { Id = 2, Name = "Meena", Salary = 70000 };
            Employee emp3 = new Employee { Id = 3, Name = "Karan", Salary = 65000 };

```

```

    IEmployeeRepo<Employee> empRepo = new EmployeeRepo<Employee>();
    empRepo.AddEmployee(emp1);
    empRepo.AddEmployee(emp2);
    empRepo.AddEmployee(emp3);

    foreach (var emp in empRepo.GetAllEmployees())
    {
        Console.WriteLine(emp);
    }

    Console.WriteLine(" ");

    Manager mgr1 = new Manager { Id = 101, Name = "Sunil", Salary = 90000, Department
= "HR" };
    Manager mgr2 = new Manager { Id = 102, Name = "Priya", Salary = 95000, Department
= "IT" };

    IEmployeeRepo<Manager> mgrRepo = new EmployeeRepo<Manager>();
    mgrRepo.AddEmployee(mgr1);
    mgrRepo.AddEmployee(mgr2);

    foreach (var mgr in mgrRepo.GetAllEmployees())
    {
        Console.WriteLine(mgr);
    }

    Console.WriteLine(" ");

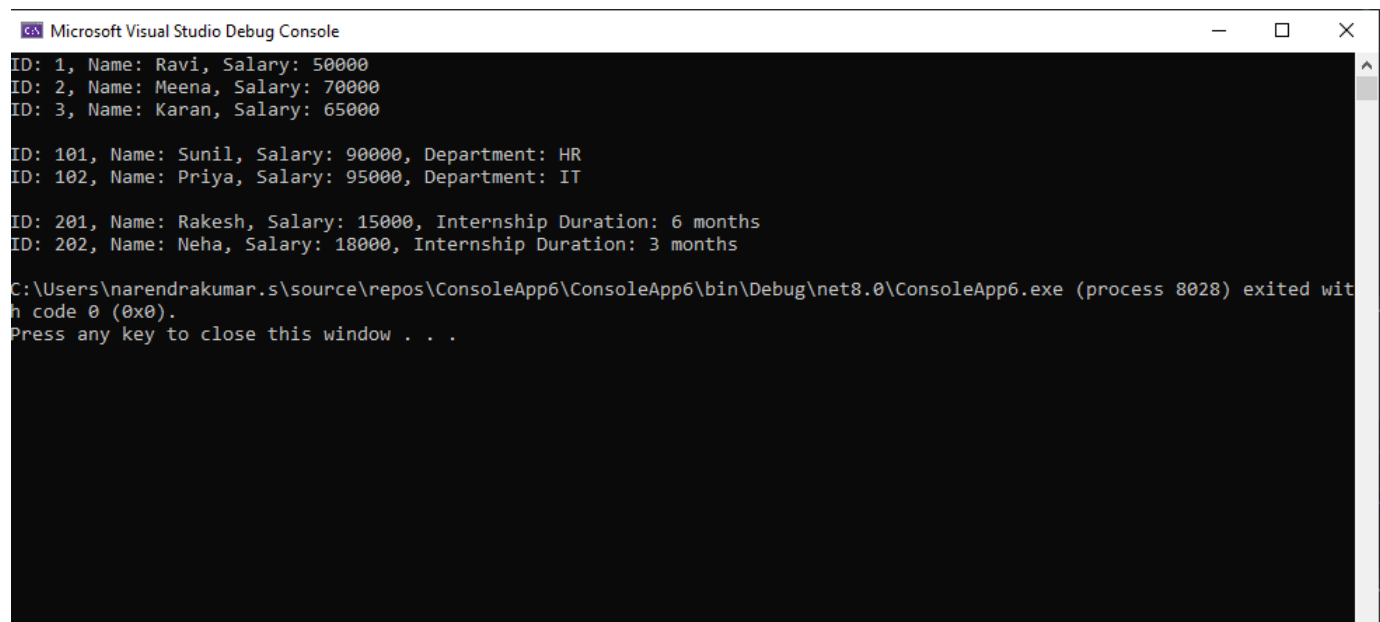
    Intern intern1 = new Intern { Id = 201, Name = "Rakesh", Salary = 15000,
DurationInMonths = 6 };
    Intern intern2 = new Intern { Id = 202, Name = "Neha", Salary = 18000,
DurationInMonths = 3 };

    IEmployeeRepo<Intern> internRepo = new EmployeeRepo<Intern>();
    internRepo.AddEmployee(intern1);
    internRepo.AddEmployee(intern2);

    foreach (var intern in internRepo.GetAllEmployees())
    {
        Console.WriteLine(intern);
    }
}
}
}

```

Output :

A screenshot of the Microsoft Visual Studio Debug Console window. The window has a title bar with the text "Microsoft Visual Studio Debug Console" and standard minimize, maximize, and close buttons. The console area is black with white text. The output shows several lines of data: three employee records (ID, Name, Salary), two records with department information (ID, Name, Salary, Department), and two records with internship duration (ID, Name, Salary, Internship Duration). The program has exited with code 0, and a prompt asks the user to press any key to close the window.

```
Microsoft Visual Studio Debug Console

ID: 1, Name: Ravi, Salary: 50000
ID: 2, Name: Meena, Salary: 70000
ID: 3, Name: Karan, Salary: 65000

ID: 101, Name: Sunil, Salary: 90000, Department: HR
ID: 102, Name: Priya, Salary: 95000, Department: IT

ID: 201, Name: Rakesh, Salary: 15000, Internship Duration: 6 months
ID: 202, Name: Neha, Salary: 18000, Internship Duration: 3 months

C:\Users\narendrakumar.s\source\repos\ConsoleApp6\ConsoleApp6\bin\Debug\net8.0\ConsoleApp6.exe (process 8028) exited with code 0 (0x0).
Press any key to close this window . . .
```